	LLaMEA (mean ± std over 5 runs)			Difference between arms		Baselines (15 seeds)		
Problem	**Penalty**	**Free**	**Informed v2**	**Free − Penalty**	**Informed v2 − Free**	**COLSHADE**	**L-SHADE + Deb**	**L-SHADE + penalty**
lens/original	—	**0.685**	0.616	—	-0.069	0.582	0.586	0.585
lens/realistic_draft	—	**0.671**	0.640	—	-0.031	0.579	0.582	0.587
cec/RC12	—	**0.794**	0.788	—	-0.006	0.643	0.621	0.512
cec/RC13	—	0.959	**0.961**	—	+0.002	0.849	0.795	0.799
cec/RC15	—	0.944	**0.949**	—	+0.006	0.711	0.696	0.699
cec/RC16	—	**0.588**	0.496	—	-0.092	0.486	0.496	0.494
cec/RC18	—	0.615	**0.653**	—	+0.038	0.591	0.606	0.645
cec/RC21	—	0.798	**0.917**	—	+0.119	0.811	0.743	0.738
cec/RC25	—	**0.561**	0.535	—	-0.026	0.515	0.514	0.506
cec/RC26	—	0.499	**0.510**	—	+0.011	0.375	0.427	0.446
Mean over problems	—	**0.711**	0.707	—	-0.005	0.614	0.607	0.601

Test AOCC (higher is better; ≥ 0.5 means feasible). LLaMEA: mean ± std over runs (Penalty n=0, Free n=1, Informed v2 n=1) of the mean over 15 test seeds of each run's best algorithm. Bold: best
Difference: of the means; δ = Cliff's delta over runs; * Mann-Whitney p < 0.05 (two-sided, uncorrected). Tint: blue = first arm better, red = worse.